Thermodynamics of Locking: The Boundary Decided

Author: Hadi Servat

Affiliation: Independent Researcher

Contact: publish@fractalresonance.com

Website: fractalresonance.com

DOI: 10.5281/zenodo.21264195 · **Version:** v2.1
Permanent record: https://doi.org/10.5281/zenodo.21264195 · PDF: https://zenodo.org/records/21264195

Abstract

> *Version 2.1 records the decision of the boundary probe that v2.0 registered as its decisive measurement. The question: does the combined ledger form sigma_566 = sigma_ST + k d ln C remain non-negative on the unlock side — equivalently, does destroying coherence cost at least k of entropy production per nat of log-coherence lost (the Landauer-flavored bound)? The upgraded instrument eliminates the v2.0 systematic floor by construction: the model's free energy is exact by quadrature, so sigma_ST = (W − ΔF_exact)/T with exact per-walker SEM. Result, seed-replicated: sigma_ST falls monotonically through the bound requirement (0.1322 nats, exact) with no plateau — sigma_566 = −0.0659 ± 0.0017 at T_down = 96 and −0.1034 ± 0.0012 at*

> *T_down = 192, invariant under every k* > 0. The strong bound is killed in the instrumented model class. The conceptual root: the Landauer analogy inverted a sign — bit erasure lowers system entropy (a locking operation, where positivity holds); coherence destruction raises it, so the system pays its own ledger and the bath owes nothing. The combined ledger is directional: locking-side positivity is implied by standard thermodynamics and was instrument-verified term by term (first drive-and-bath closure in the corpus, dQ = sigma − dS_env closing numerically). Corrections cascade: FRC 566.001 §5.3 is withdrawn by companion note; FRC 566.030 restates its inequality as directional in v1.1, its accounting prediction VERIFIED by this same instrument. Predictions P-T1/P-T2 are unchanged and live.*

1. What this version is

v2.0 (published 2026-07-08) rebuilt this paper from a consistency sketch into an instrument, isolated the ledger's only discriminating content on the unlock side, built the probe, and scored it INCONCLUSIVE at its systematic floor, registering the precision upgrade as the next step. v2.1 executes that registered step and records the decision. Nothing else in v2.0 is retracted: Model A's locking results, Model B's record-channel results, the Gate-3 verdict template, and predictions P-T1/P-T2 stand as published, with their priority.

This revision is filed under the corpus rule for weak or decided core claims: update, do not bury. The negative result is kept at full volume (metadata: results-negative-kept). An independent audit document prepared for adversarial model review, FRC 828.001, accompanies this revision and addresses the two historical false-kill patterns (closed-system misreading; constant-k misreading) explicitly.

Lineage: v1 (consistency sketch) 10.5281/zenodo.17438231; v2.0 10.5281/zenodo.21254142; this version 10.5281/zenodo.21264195.

2. The question, as v2.0 registered it

Write sigma_566 = sigma_ST + k*·d ln C, with sigma_ST = dS_sys + dS_env the standard entropy production. v2.0 §3 established the asymmetry:

- **Locking (d ln C > 0):** sigma_566 ≥ 0 is implied by standard thermodynamics. Bookkeeping. Verified by instrument in v2.0 (A2).

- **Unlocking (d ln C < 0):** sigma_566 ≥ 0 asserts sigma_ST ≥ k*·|d ln C| — a Landauer-flavored floor on coherence erasure that standard thermodynamics does not guarantee. This was the paper's registered decisive measurement, and the strongest single result the 566 line could have owned.

3. The upgraded instrument

The v2.0 floor came from measuring sigma_ST through two bias-accumulating channels (histogram entropy plus Stratonovich heat integration). The v2.1 instrument removes both from the primary channel:

1. **Exact free energy.** For Model A (unchanged: overdamped Langevin phases, U(φ,t) = −A(t)·ln C(φ), three-basin landscape at weights 0.66/0.24/0.10, β = 4, T = 1, k* = 1 nat), the partition function Z(A) = ∫C(φ)^{A/T}dφ is computed by quadrature on a 4×10^5 grid. F(A) is exact. For an isothermal protocol between equilibrated endpoints, sigma_ST = (W − ΔF_exact)/T.
2. **Exact statistics.** Work is accumulated per walker (Jarzynski discrete convention); walkers are independent, so SEM(W) = std(w_i)/√N.
3. **Exact start, verified end.** Initial ensembles are drawn from the exact equilibrium at A = 6 by inverse-CDF sampling; final ensembles are verified against exact quadrature values (S within 0.006, U within 0.003, C_lock within 0.003 in all runs).
4. **Audits.** Jarzynski identity closes against exact ΔF; the v2.0 two-channel instrument, run alongside at T_down = 12, agrees with the new channel within 0.005; sigma_ST > 0 in every window (the theorem the instrument must obey).

Exact anchors: ΔF(unlock, 6 → 0.5) = 17.885932; Δln C_lock (quasi-static) = −0.132245. **Bound requirement: 0.132245 nats.**

4. The decision

Unlock protocol A = 6 → 0.5 over T_down, then 10 units of relaxation; N = 48,000; dt = 0.002; fixed seeds; seed-replicated at the decisive points.

T_down	seeds	sigma_ST (mean ± seed scatter)	sigma_566
12	1	0.3586 ± 0.0028	+0.2260
24	3	0.2304 ± 0.0052	+0.0982
48	3	0.1287 ± 0.0048	−0.0035 (crossing)
96	3	0.0663 ± 0.0017	**−0.0659**
192	2	0.0288 ± 0.0012	**−0.1034**

Controls on the number that decides:

- **Discretization:** three seeds per dt at T_down = 24 give sigma_ST means 0.2156/0.2304/0.2327 for dt = 0.004/0.002/0.001; Richardson extrapolation bounds the dt = 0.002 bias at ≲ +0.005 (upward). Applied in full at T_down = 192 the verdict moves to −0.098. The sign cannot be rescued.
- **No plateau:** sigma_ST decays through the 0.1322 requirement between T_down = 24 and 48 and keeps falling; between 96 and 192 the local decay exponent is ≈ 1.2 (faster than 1/T). The quasi-static limit of sigma_566 is k*·Δln C = −0.1322 exactly.
- **k\-*invariance:*** *Δln C is a state-function difference fixed by the endpoints; sigma_ST → 0 is measured. Hence sigma_566 → k\·*(−0.1322) < 0 for every k* > 0. No choice of the units bridge changes the verdict; only a state-dependent (i.e., fitted) k could, and the corpus forbids that reading ([[FRC-566-030]] §3).

Verdict: the strong unlock-side bound fails in the instrumented model class. Destroying coherence quasi-statically costs asymptotically nothing; the combined form sigma_566 ≥ 0 is directional, with content only on the locking side, where it is implied by standard thermodynamics.

5. Why the bound was never available (recorded interpretation)

The Landauer analogy that motivated the bound inverts a sign. Landauer erasure (RESET of a memory) *lowers* the system's entropy, so the bath must absorb the difference and the cost survives the quasi-static limit. Coherence destruction *raises* the system's entropy — the ensemble spreads from locked basins toward the flat landscape (measured here: ΔS_sys = +1.4 on slow unlock, with heat flowing from bath *into* system). The system's own entropy rise pays the combined ledger; nothing forces an export. In FRC's own vocabulary, the true analogue of Landauer's RESET is **locking** — and there the positivity holds, is standard, and was verified in v2.0. Nature charges admission to order; it does not charge exit.

6. What this paper now is

100.005's role in the series was never a discovery claim; it is the license and the court — the paper that certifies which thermodynamic sentences the corpus may utter. In that role, v2.0 + v2.1 delivers:

1. **The instrumented locking closure** (v2.0 A2): first drive-and-bath closure demanded by [[FRC-566-030]] §7 — dQ = sigma − dS_env closing numerically with every term independently measured. The accounting half of 566.030's registered prediction is thereby VERIFIED.

2. **The decided boundary** (this version): the positivity half of that same prediction is REFUTED on the unlock side; [[FRC-566-001]] §5.3 is withdrawn by companion correction note; [[FRC-566-030]] v1.1 restates its §6 inequality as directional.
3. **The reference Gate-3 implementation**: the exact-ΔF route is promoted as the standard instrument for auditing any FRC collapse model's entropy bill ([[FRC-100-003]] v3 Gate 3). The v2.0 verdict template gains step (vii): state the unlock-side status against the exact-anchor instrument.
4. **Live discriminating predictions**: P-T1 (dissipation bursts concurrent with locking) and P-T2 (trajectory correlation, measured r = +0.61 in v2.0) are untouched and remain the paper's experimental face, pointed at Minev-class records.

7. Successor claim — registered, NOT claimed

With the quasi-static floor dead, the physically operative question moves to finite time — where FRC measurement actually lives ([[FRC-100-003]]: collapse is finite-time locking). Registered: the finite-time unlock cost coefficient c(T_down) = sigma_ST·T_down, whose convergence would give a thermodynamic-length-like geometry of the coherence landscape. Current data: c = 4.3, 5.5, 6.2, 6.4, 5.5 across T_down = 12–192 — **non-monotone, not converged, and therefore not claimed**. Deciding its asymptote (or showing there is none in this barrier-dominated landscape) is appendix work with a pre-registered protocol required first.

8. Kill conditions (this version's own)

The v2.1 decision fails if: the quadrature anchors are wrong (checkable by any independent integrator); the identity sigma_ST = (W − ΔF)/T is misapplied (requires non-equilibrated endpoints — excluded by the endpoint audits); the dt bias exceeds its Richardson bound by an order of magnitude; or seed replication at N ≥ 10× fails to reproduce the sign at T_down ≥ 96. Each is a checkable, physical failure mode; none is currently indicated.

9. Appendix program (100.005.001, updated)

1. ~~Precision upgrade and boundary decision~~ — **DONE, this version.**
2. Explicit-bath model (collision/spin-boson) with I(sys:env) instrumented — unchanged.
3. FDT and detailed-balance checks — unchanged.
4. Protocol-efficiency study across measurement strengths — unchanged.
5. P-T1 against Minev-class catch-and-reverse records, markers pre-registered first — unchanged, now the paper's primary experimental target.
6. NEW: pre-registered protocol for the finite-time coefficient c(T_down) (§7), including barrier-height sweeps to separate landscape geometry from Kramers effects.

Connections

- [[FRC-566-001]] — Reciprocity law and UCC; §5.3 Landauer-coherence bound withdrawn by companion note
- [[FRC-566-030]] — Open-system ledger; accounting half verified, positivity restated as directional (v1.1)
- [[FRC-100-003]] — Collapse as finite-time locking; this paper supplies the reference Gate-3 instrument
- [[FRC-100-006]] — Born-weighted landscape used by Model A
- [[FRC-787-787]] — Lambda-flight transport
- FRC 828.001 (adversarial audit, corpus record) — Independent audit document (adversarial review)

References

1. K. Sekimoto, *Stochastic Energetics*, Springer (2010).
2. U. Seifert, "Stochastic thermodynamics, fluctuation theorems and molecular machines," *Rep. Prog. Phys.* 75, 126001 (2012).
3. R. Landauer, *IBM J. Res. Dev.* 5, 183 (1961).
4. T. Sagawa, M. Ueda, *PRL* 100, 080403 (2008).
5. C. Jarzynski, *PRL* 78, 2690 (1997).
6. G. E. Crooks, *Phys. Rev. E* 60, 2721 (1999).
7. D. A. Sivak, G. E. Crooks, "Thermodynamic metrics and optimal paths," *PRL* 108, 190602 (2012).
8. Z. K. Minev et al., *Nature* 570, 200 (2019).
9. H. Servat, *FRC 100.005 v2.0 — Thermodynamics of Locking.* DOI: 10.5281/zenodo.21254142.
10. H. Servat, *FRC 100.003 v3.1 — Collapse as Open-System Phase-Locking.* DOI: 10.5281/zenodo.21253398.
11. H. Servat, *FRC 566.001 — Entropy–Coherence Reciprocity and UCC* (+ 2026-07-08 correction note to §5.3). DOI: 10.5281/zenodo.21264217.
12. H. Servat, *FRC 566.030 v1.1 — Reciprocity in Action* (directional restatement). DOI: 10.5281/zenodo.21264213.
13. H. Servat, *FRC 828.001 — Anatomy of a Kill* (audit document, 2026-07-08).